Problem Tracker — Segment 2: Live Demo

2 of 8 · 5 min

Grounding the architecture discussion in something working before you explain it

> **Transition in:** "Here's the actual app — I'll click through the four tabs that matter, then we'll get into how it's built."

Before You Start

- Launch from the project's own venv, not a global interpreter: `source venv/bin/activate && streamlit run dashboard.py`
- Confirm the browser tab is actually pointed at that server — a stray global Streamlit install (1.12.0, from 2022) caused two real bugs recently; know that story if it comes up.
- Have run `tracker.py` recently so Recommended and Statistics have real, fresh data.
- Number to have ready: **354 problems tracked** across NeetCode and LeetCode.

The 5-Minute Run of Show

0:00–1:00	Today's Queue — the core loop
1:00–2:00	Recommended — a real schema change to build
2:00–2:45	Overdue — a real bug-fix story
2:45–3:30	Browse + Statistics — quick pass
3:30–4:15	Bonus: the daily email digest
4:15–5:00	Close + transition to architecture

Click-by-Click Script

Today's Queue (0:00–1:00)

- Drag the capacity slider. *"This picks the highest-priority problems for today — a hand-rolled heuristic weighted by staleness, past struggles, and whether it's starred, capped so no single category can dominate the queue."* Point out the new "max per category" slider too — a second knob I added after realizing the cap was hardcoded.
- Open one card, point at the badges. Don't click a review button — every click writes a real row.

Recommended (1:00–2:00)

- **New problems to try** — read a reason aloud. *"This needed a real schema change — the original DB only ever stored problems I'd solved. I added a separate catalog table from a full scrape of the problem list, not just completed ones, and diff the two to find gaps."*
- **Your weakest patterns** — *"I tightened this after testing surfaced a false positive: a category with a 0% struggle rate was still getting flagged 'weak' just for meeting the attempt-count floor."*

Overdue (2:00–2:45) — bug-fix story

- Move the "Show top N overdue" slider. *"This slider didn't exist until recently. At one point, with roughly 390 tracked problems, nearly all of them qualified as overdue, and the tab rendered every single one as a full form — well over a thousand widgets on one page. It froze the browser. I found it by actually driving the app with headless Chromium instead of eyeballing it, then capped it."*

Browse + Statistics (2:45–3:30)

- Browse: search box, difficulty filter. Statistics: point at "Lifetime solves" — *"this counts distinct problems solved combining my own accepted-submission history from both platforms with in-app review clicks — not just what I've clicked through this app."*

Bonus: The Daily Email Digest (3:30–4:15)

- *"I extended this past the dashboard — there's a script that emails me 2 review picks and 1 new recommendation every morning, using the exact same selection functions the dashboard calls. It's scheduled with launchd, and it needed zero new selection logic — just reusing what already existed and adding a delivery layer on top."*

> **Don't click Solved / Hint / Failed / Skip live** — each click writes a real row to `review_history`. Look, don't submit, so your actual practice data stays clean.